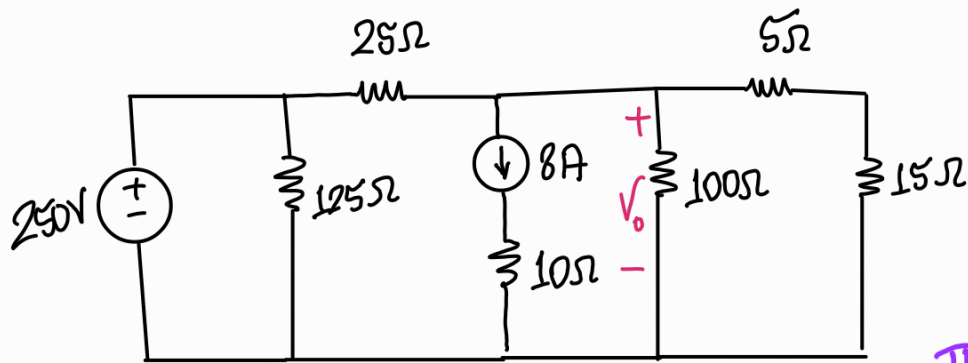
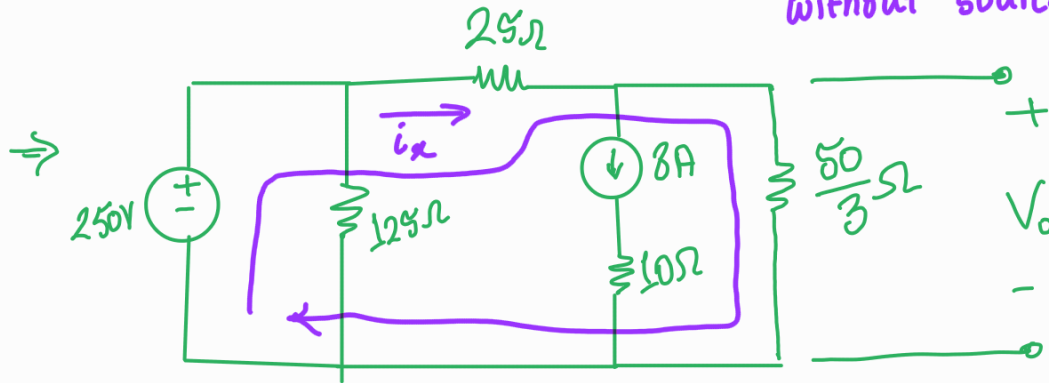


Problem 1



Find v_o and the power of the voltage source & the current source.

This problem is best solved without source transform



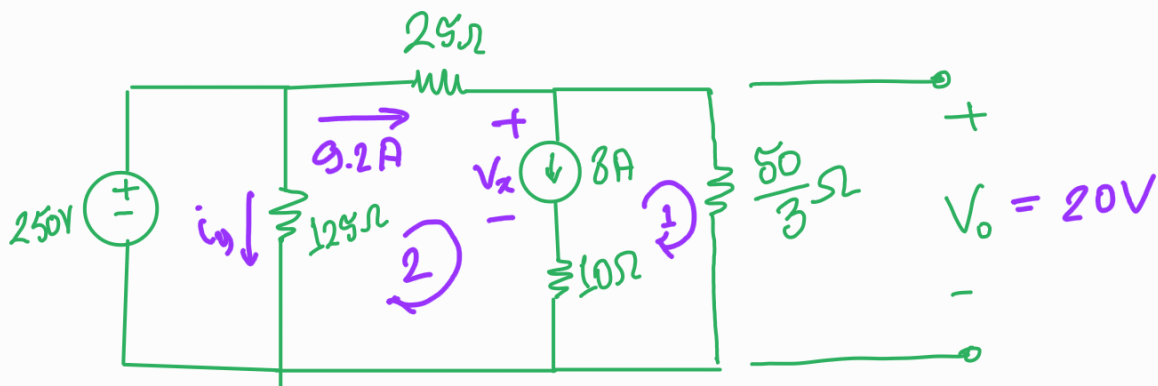
$$\text{KVL} \rightarrow -250 + 25i_x + \frac{50}{3}(i_x - 8) = 0$$

$$\Rightarrow 75i_x + 50i_x - 400 = 750$$

$$\Rightarrow 125i_x = 1150$$

$$\Rightarrow i_x = 9.2 \text{ A}$$

$$\therefore v_o = \frac{50}{3} \times (i_x - 8) = \frac{5 \times 12}{3} = 20 \text{ V}$$



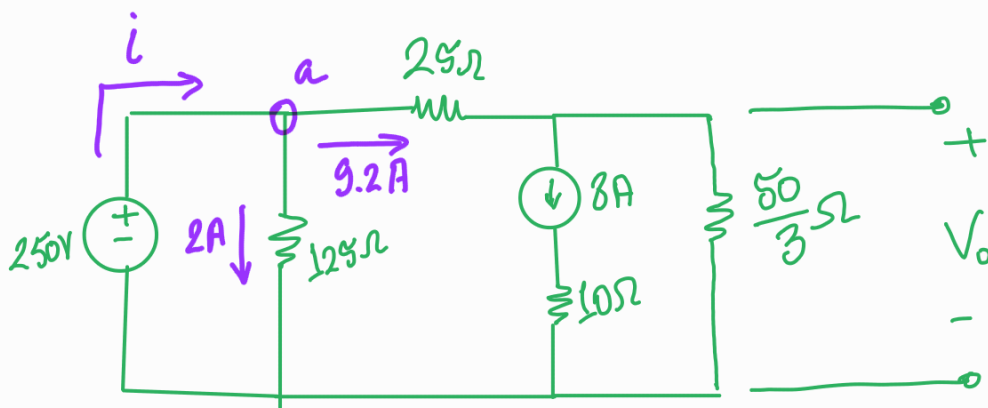
$$\text{KVL @ 1} \rightarrow -V_x + 20 - 80 = 0$$

$$\Rightarrow V_x = -60\text{V}$$

$$\therefore P_{8A} = +V_x I = (-60) \times 8 = -480\text{W} \text{ [Mistake in slide]}$$

$$\text{KVL @ 2} \rightarrow -125 i_y + 9.2 \times 25 + 20 = 0$$

$$\Rightarrow i_y = 2\text{A}$$

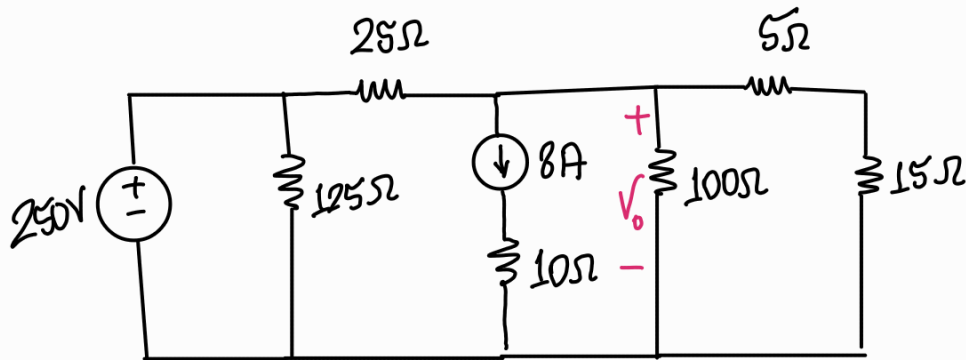


$$\text{KCL @ a} \rightarrow i = 2 + 9.2 = 11.2\text{A}$$

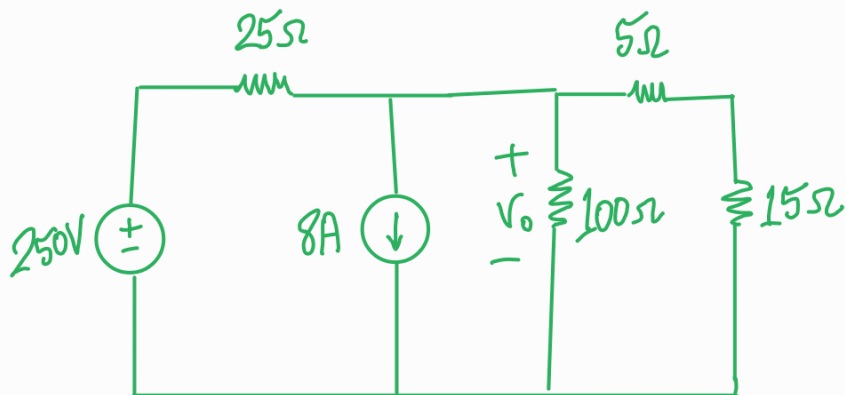
$$\therefore P_{250V} = -250 \times 11.2\text{W} = -2.8\text{KW}$$

But what if you want to solve it using source transformation?

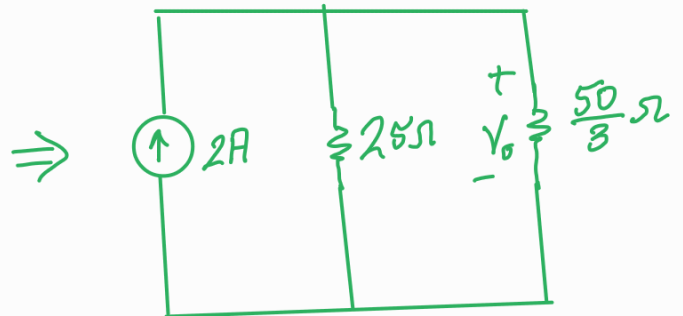
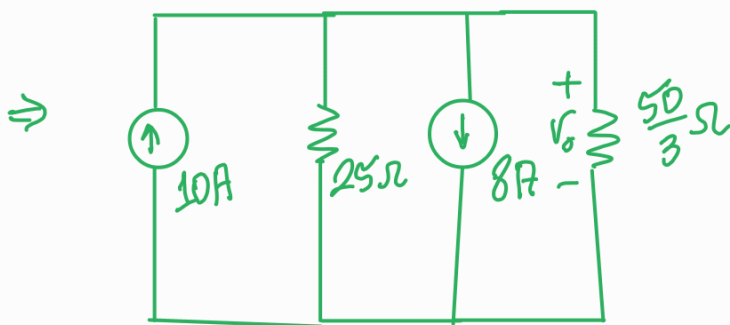
You can follow the slide based simplification. But note that you can only find V_o using the simplification. If you try finding the power in the truncated ckt, the power values will not match.



(truncation is a type of lossy compression)



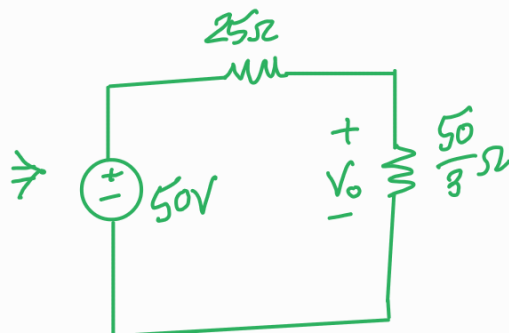
truncating following the slide's principle



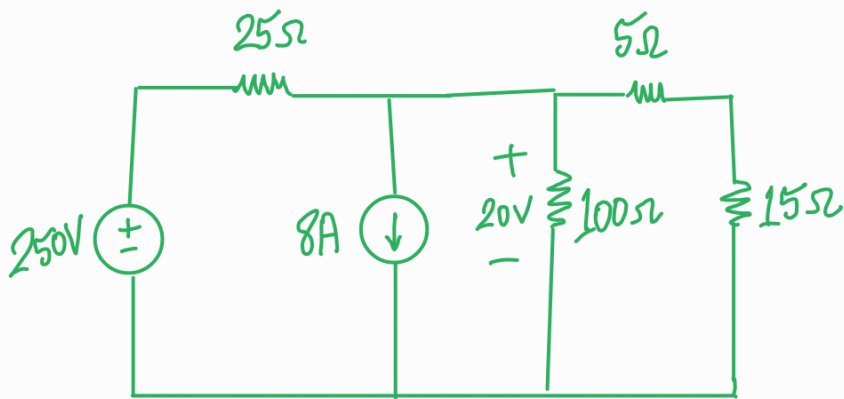
$$V_o = \frac{50/3}{25 + 50/3} \times 50 \text{ V}$$

$$= \frac{50 \times 50}{125} \text{ V}$$

= 20V ← same as KVL-KCL method

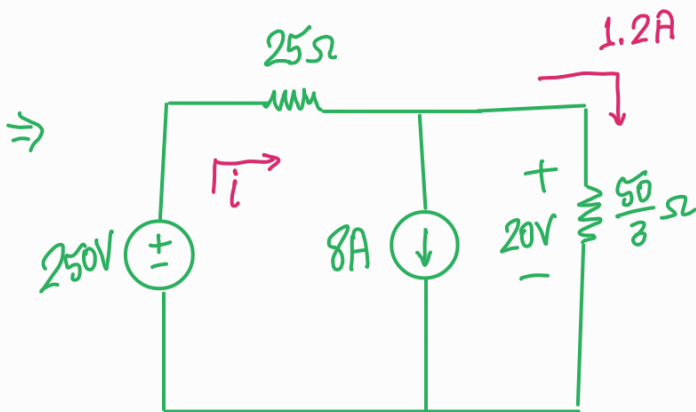


but now if you take the truncated ckt and try finding the power of the sources —



$$P_{8A}(\text{truncated}) = +20 \times 8 \text{ W} = 160 \text{ W}$$

→ doesn't match



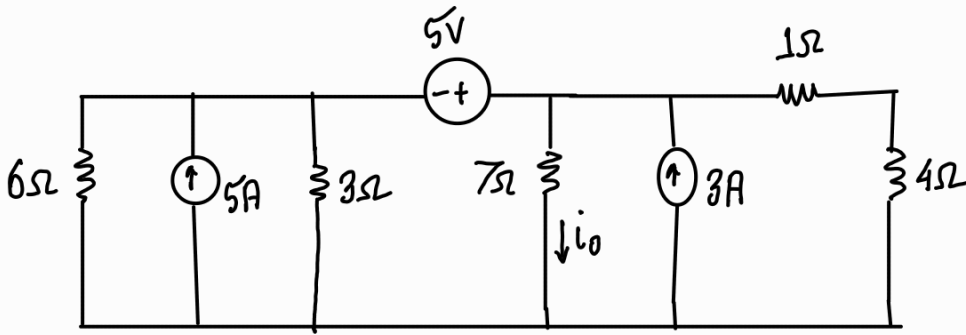
$$\Rightarrow i = 9.2 \text{ A}$$

$$\therefore P_{250V}(\text{truncated}) = -250 \times 9.2 \text{ W} = -2.3 \text{ kW}$$

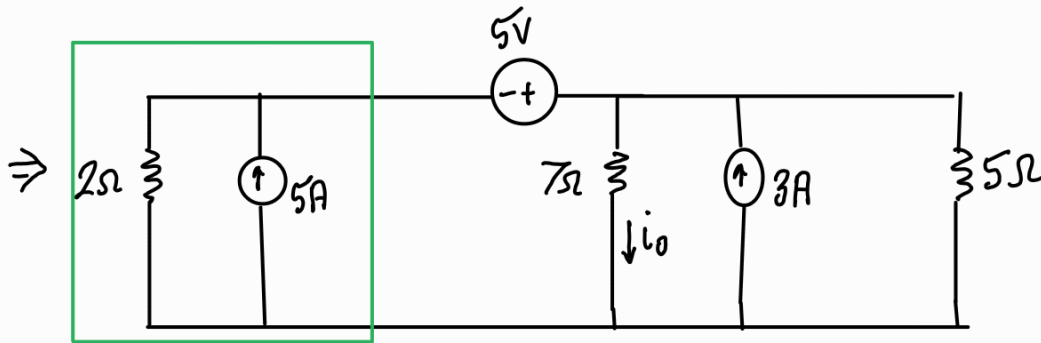
→ doesn't match

Takeaway — the chances of such a question coming in the exams is very low. Even if such a question comes you probably won't be forced to apply source transformation.

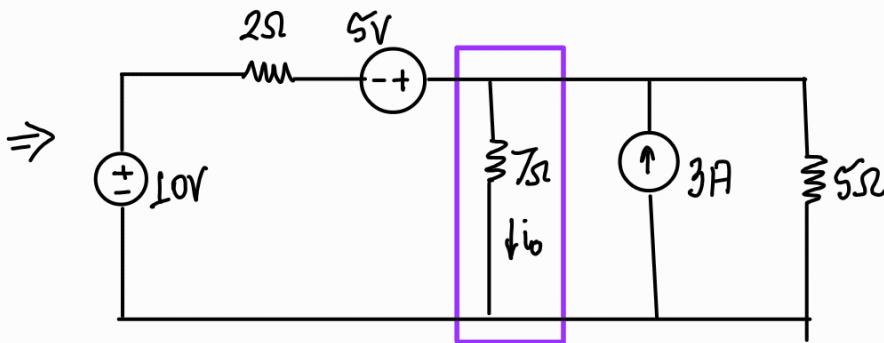
Problem 2



Find i_o using source transformation.



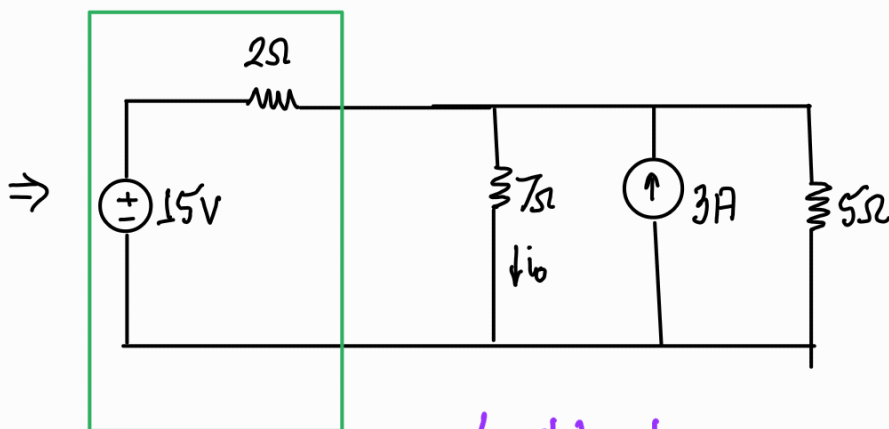
The goal of src transform is to bring the ckt in a form where you can combine resistors and sources.



target branch
(DO NOT COMBINE)

So just follow these steps —

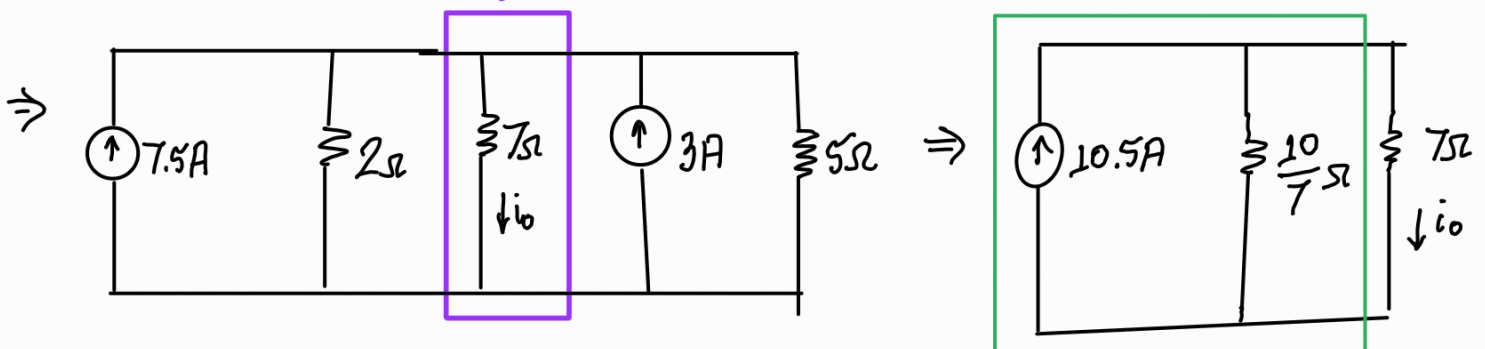
- i) simplify if possible
- ii) else apply src transform & try again

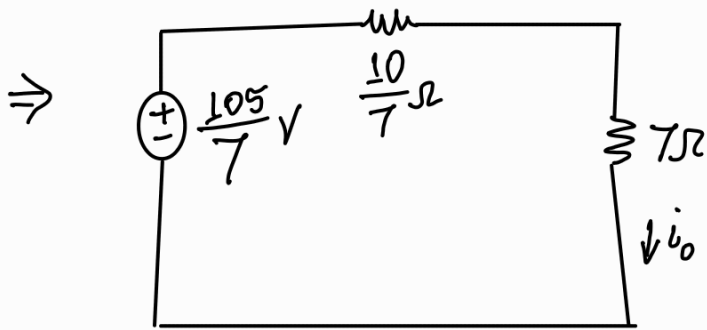


target branch

DO NOT COMBINE
TARGET BRANCHES

- iii) Continue till there is a single loop left

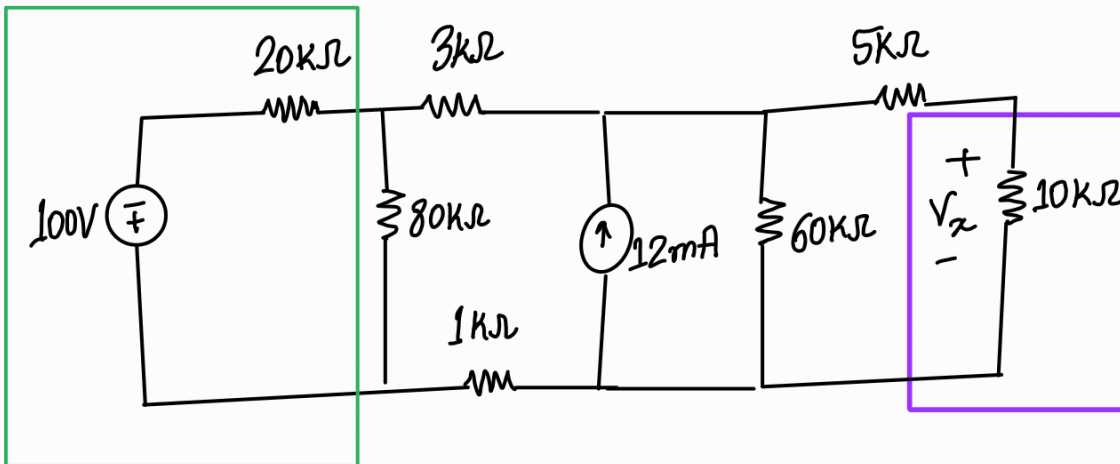




$$i_0 = \frac{\frac{105}{7}}{7 + \frac{10}{7}}$$

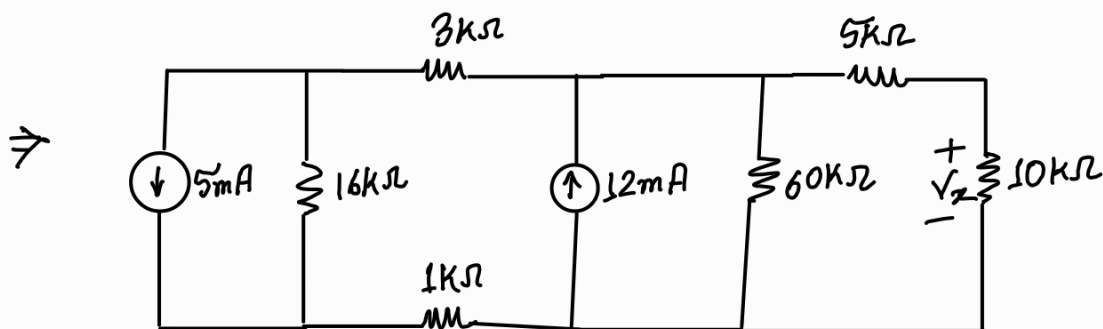
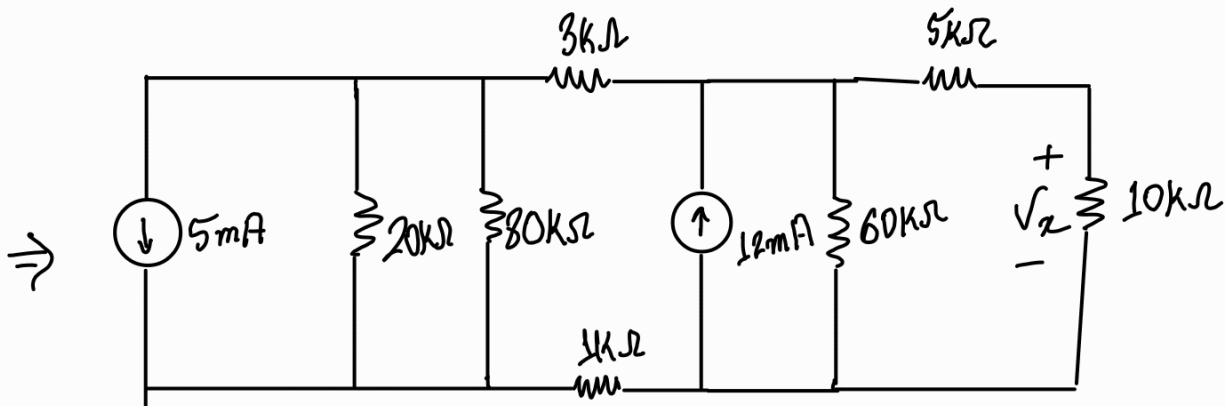
$$= \frac{105}{59} \text{ A} = 1.78 \text{ A}$$

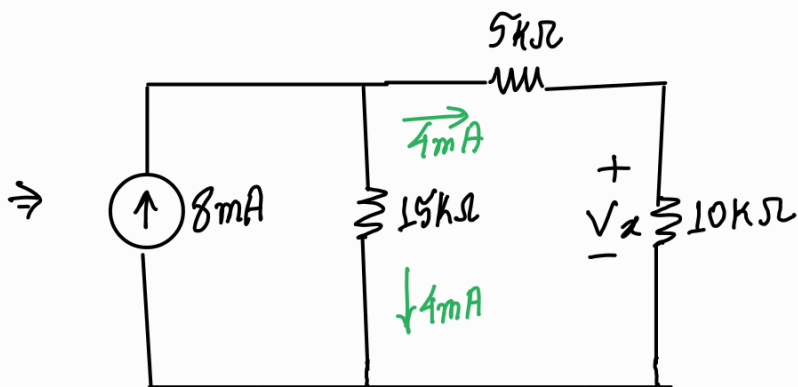
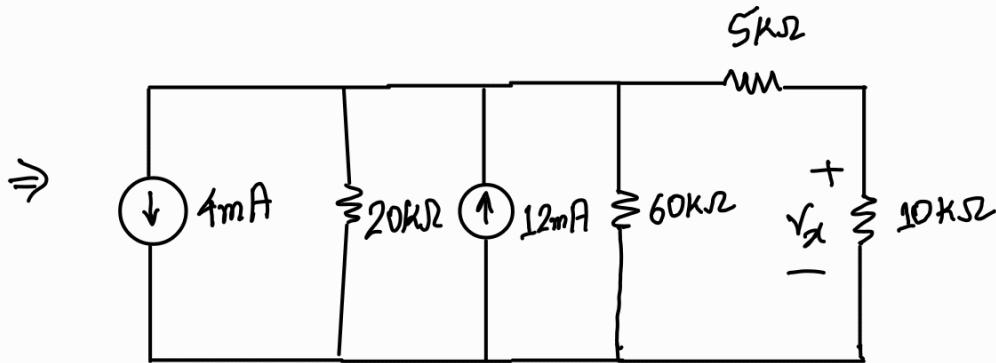
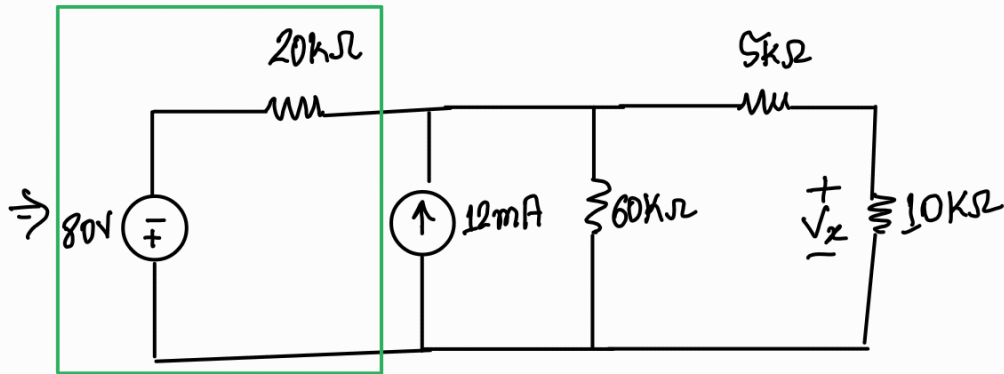
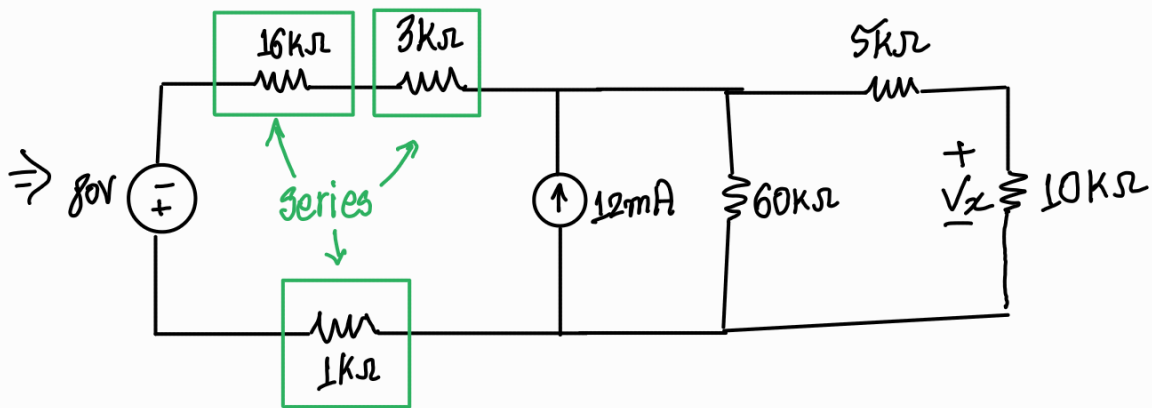
Problem 3



Find V_x

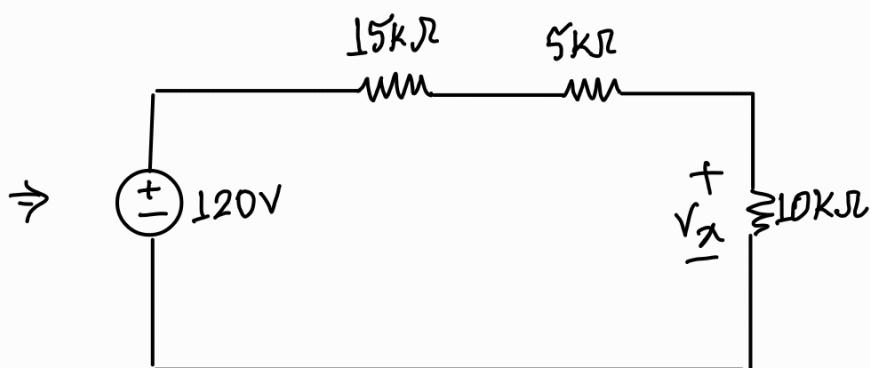
target branch





if the question doesn't ask you to bring the circuit to a single loop, you can just apply current divider here.

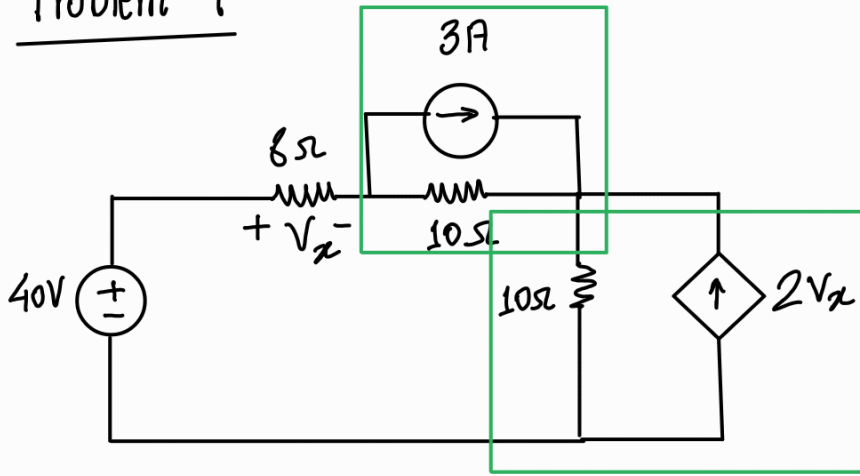
$$\therefore V_x = 4 \times 10 \text{ V} = 40\text{V}$$



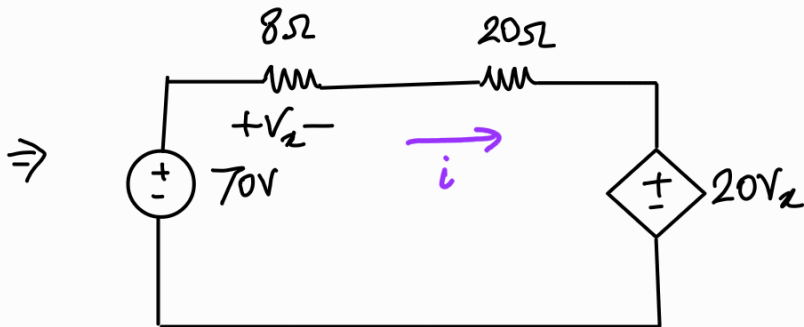
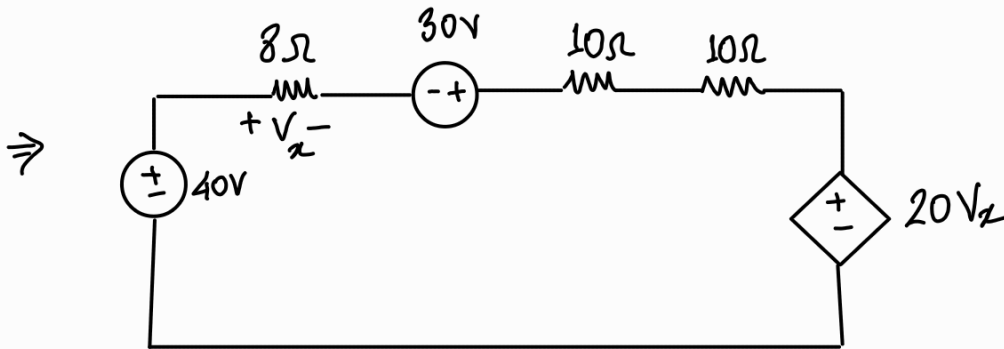
But if the question asks you to bring the ckt to a single loop, this is the form you want.

$$V_x = \frac{10}{30} \times 120 \text{ V} = 40\text{V}$$

Problem 4



Find V_x



$$i = \frac{70 - 20V_x}{8 + 20}$$

$$\text{but } i = \frac{V_x}{8}$$

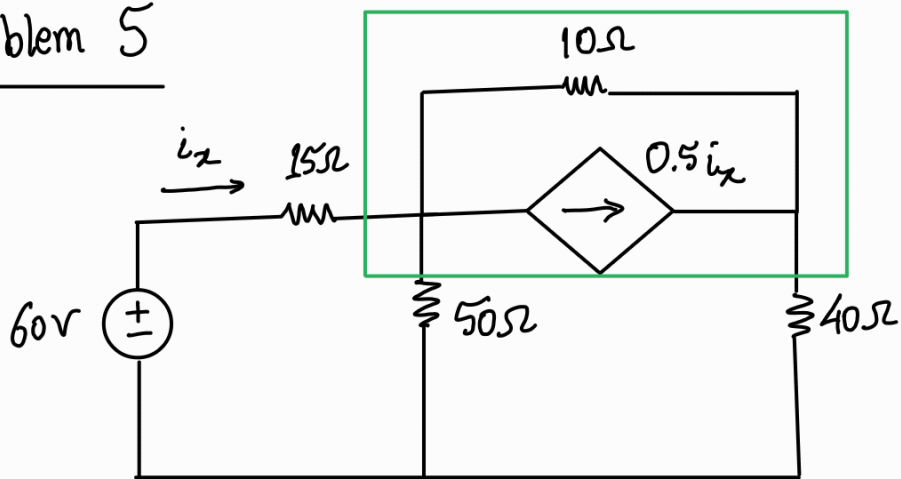
$$\therefore \frac{V_x}{8} = \frac{70 - 20V_x}{28}$$

$$\Rightarrow 28V_x = 560 - 160V_x$$

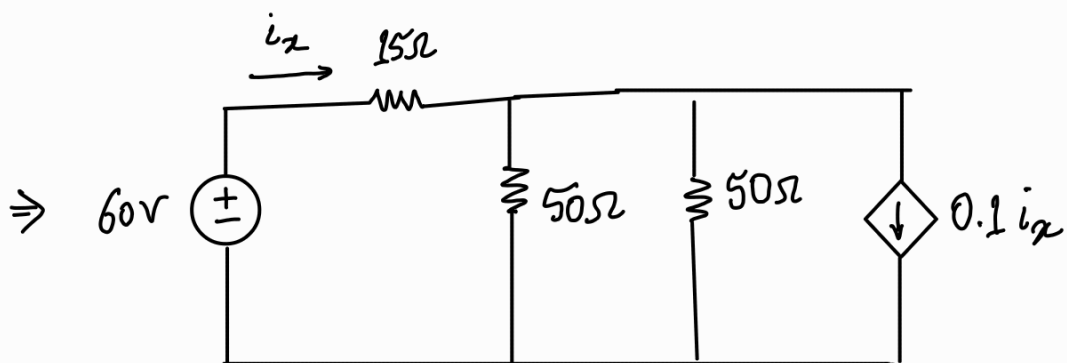
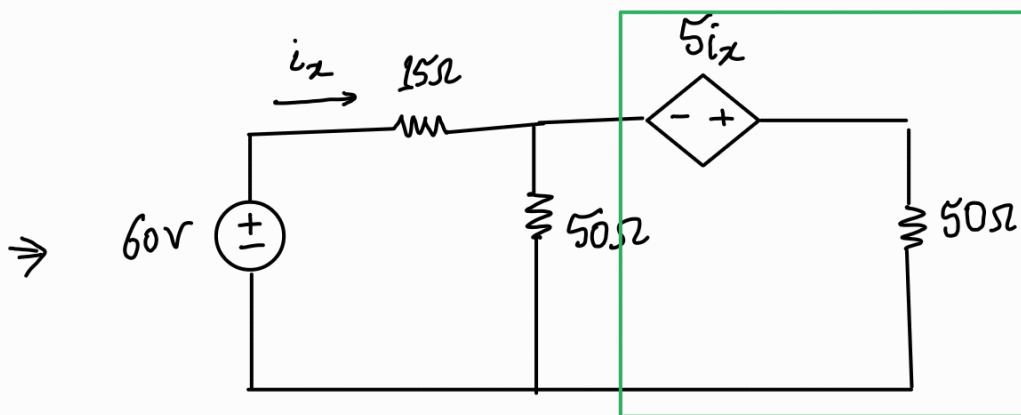
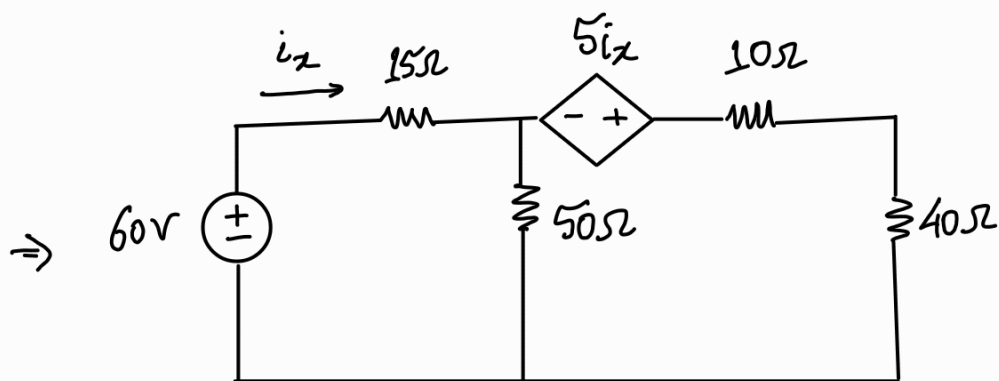
$$\Rightarrow 188V_x = 560$$

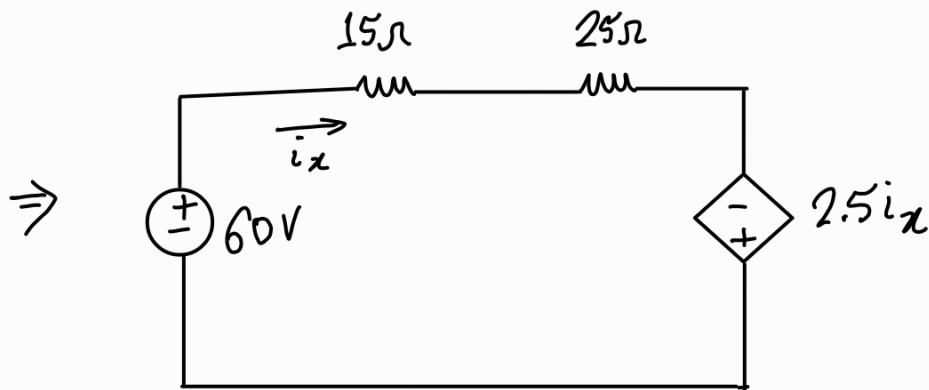
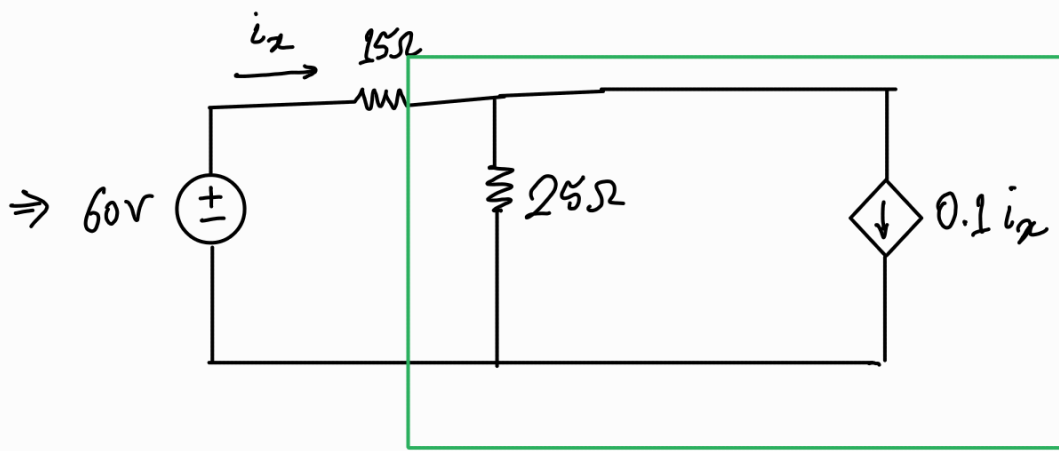
$$\Rightarrow V_x = 2.98V$$

Problem 5



Find i_x





$$i_x = \frac{60 + 2.5i_x}{40}$$

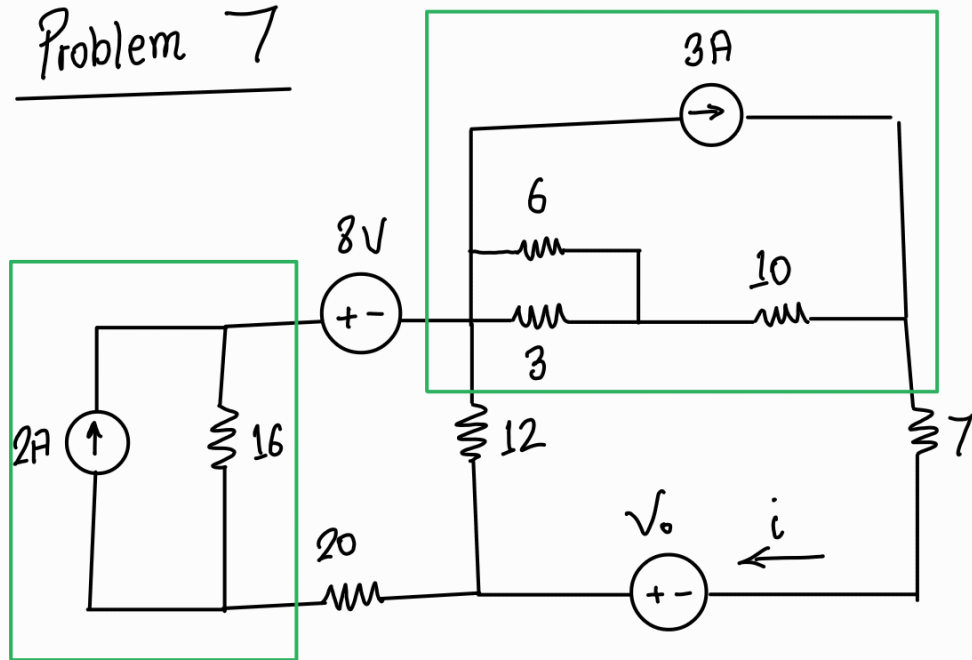
$$\Rightarrow 37.5i_x = 60$$

$$\Rightarrow i_x = 1.6 \text{ A}$$

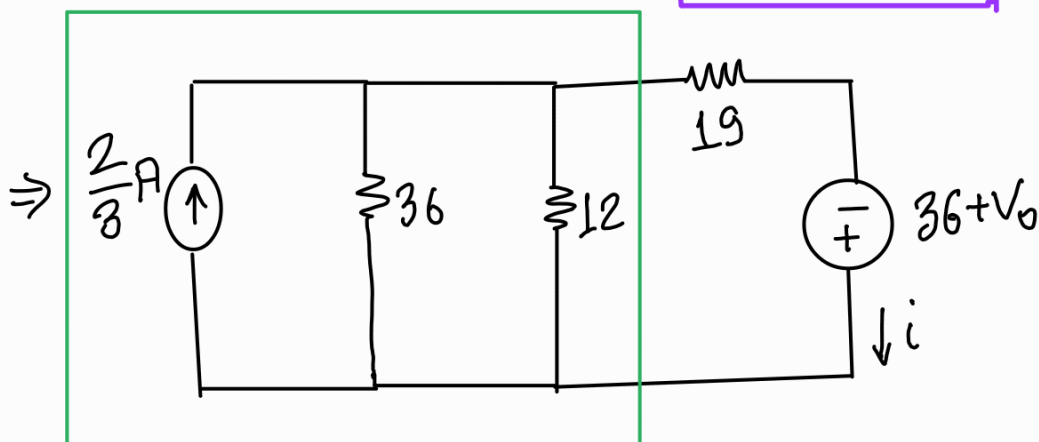
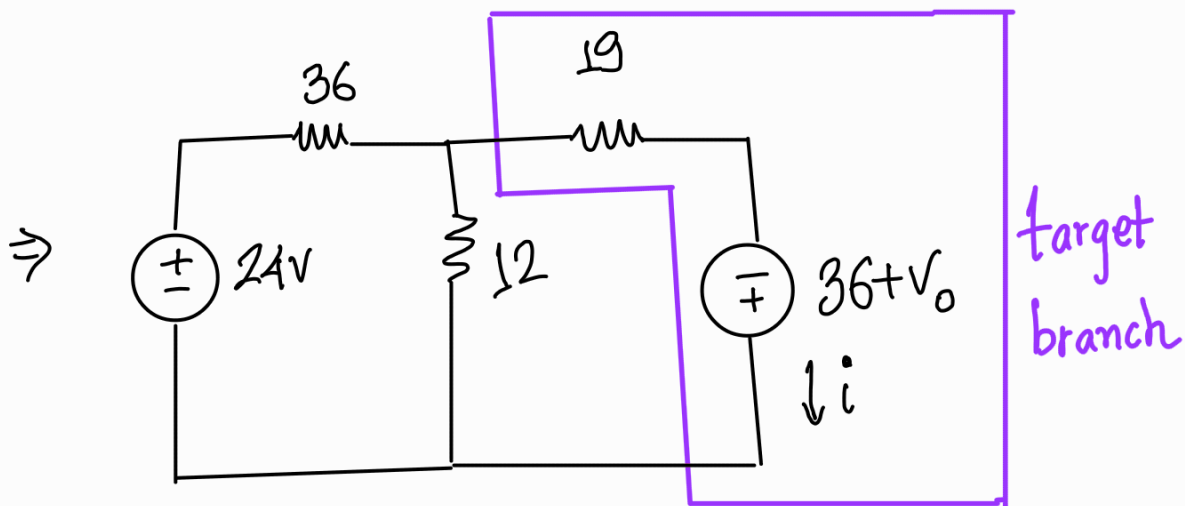
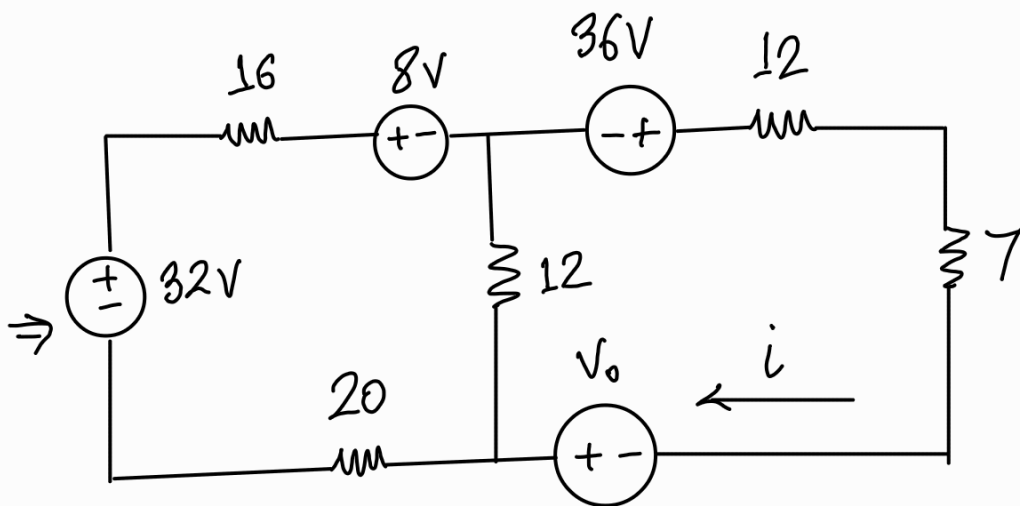
Problem 6

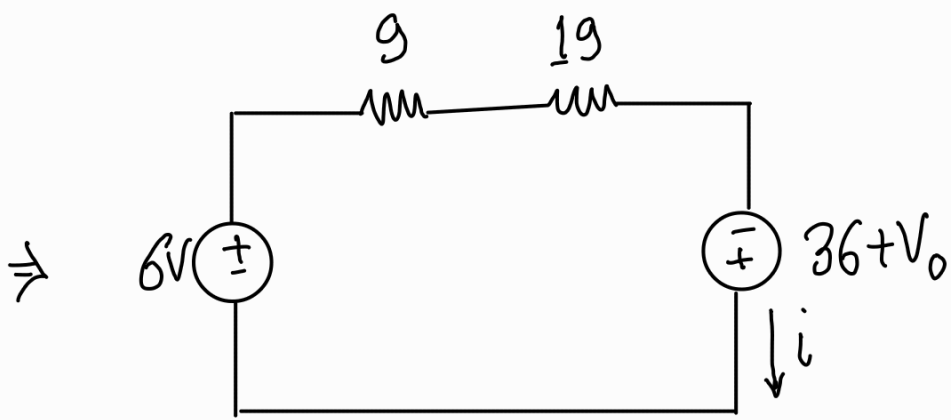
Literally the same as Problem 2.

Problem 7



Reduce the ckt to a single loop. If $i = 2.5A$ determine V_o





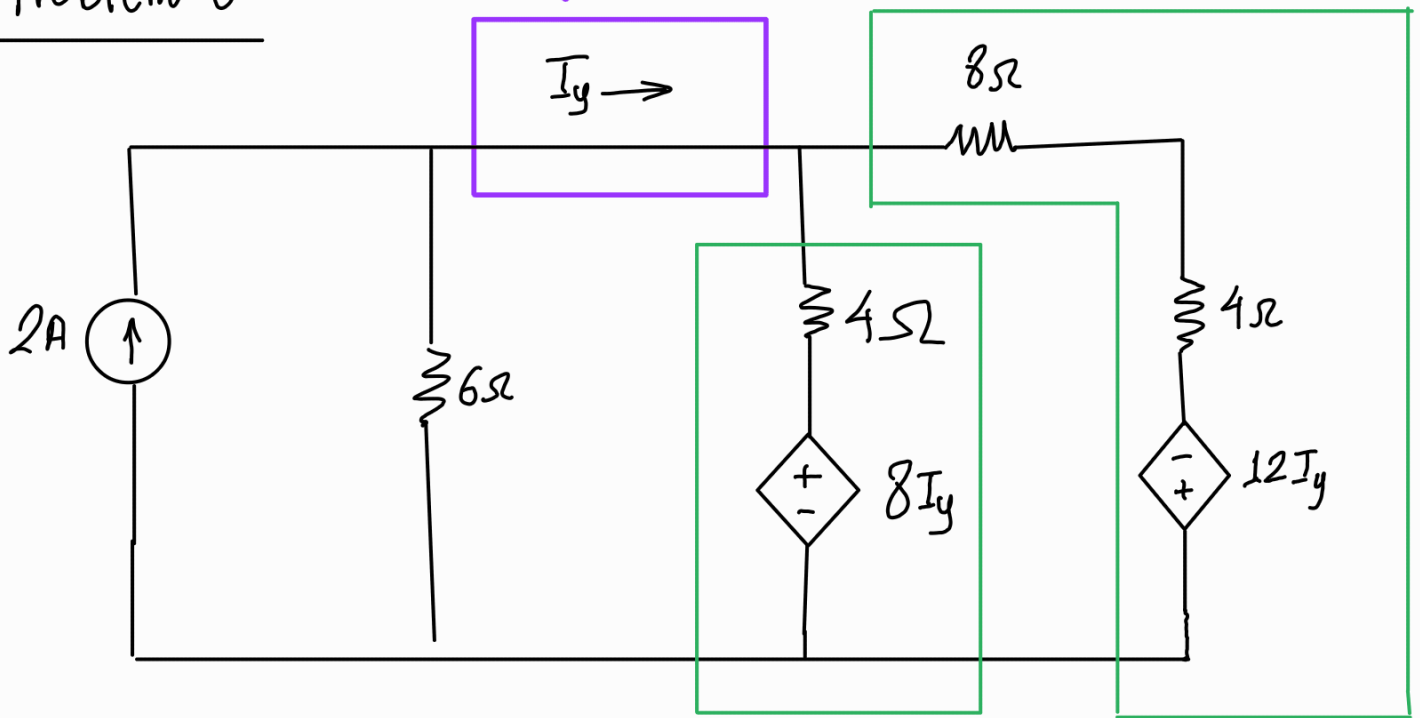
$$i = \frac{42 + V_o}{9 + 19} \Rightarrow 2.5 = \frac{42 + V_o}{28}$$

$$\Rightarrow V_o + 42 = 70$$

$$\Rightarrow V_o = 28 \text{ V}$$

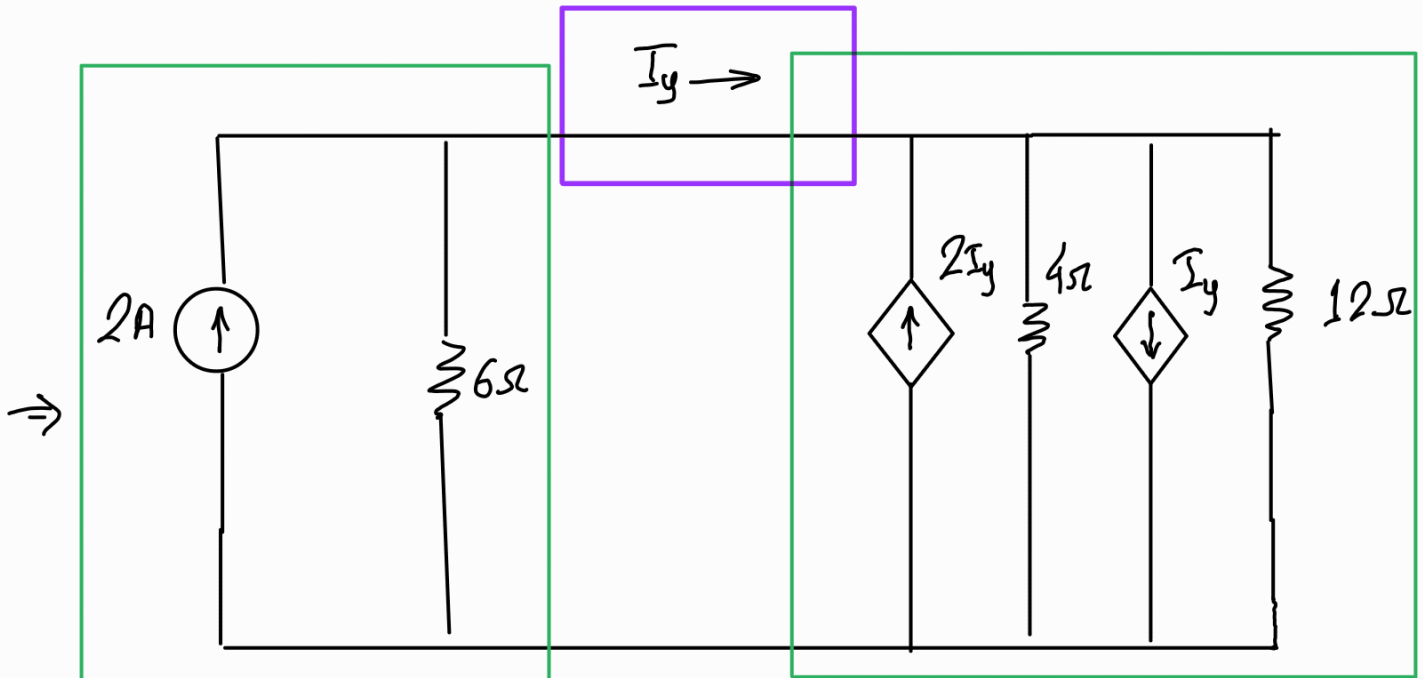
Problem 8

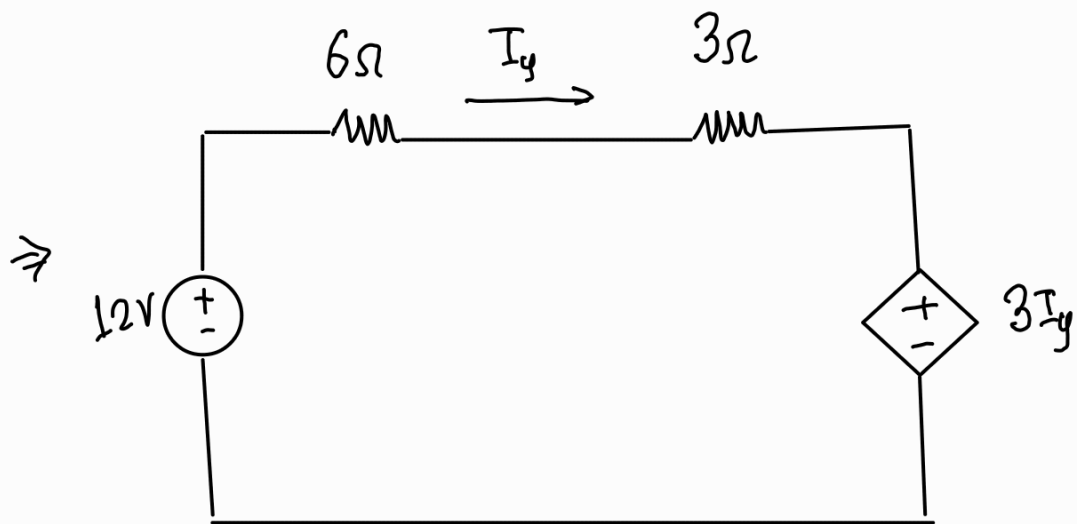
target branch



Reduce the ckt to a single loop and solve for I_y

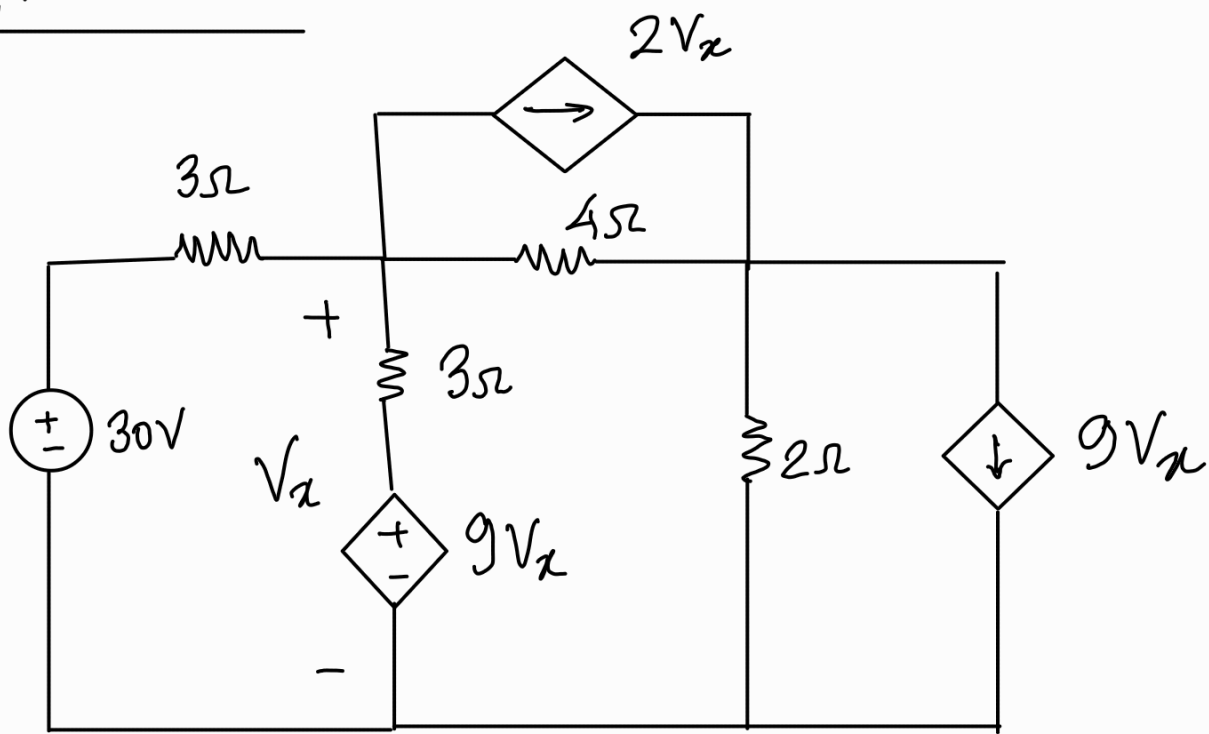
target branch



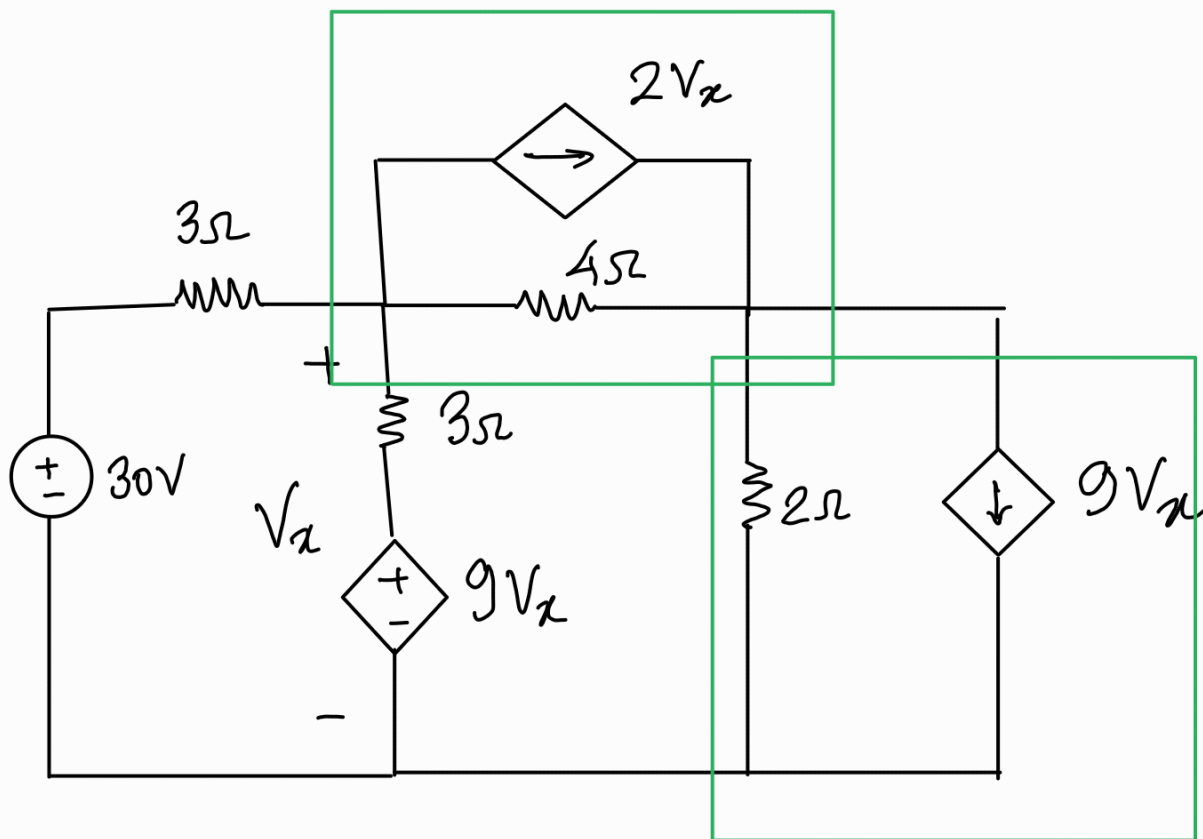


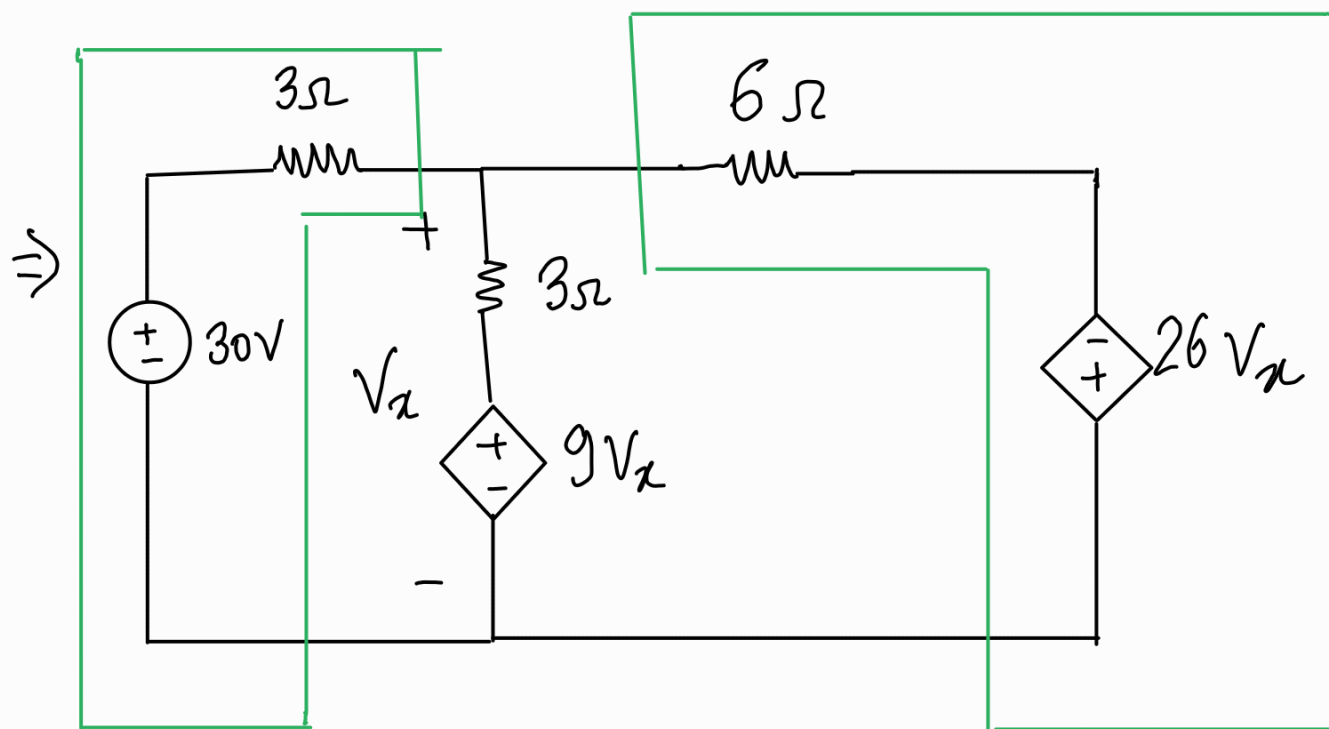
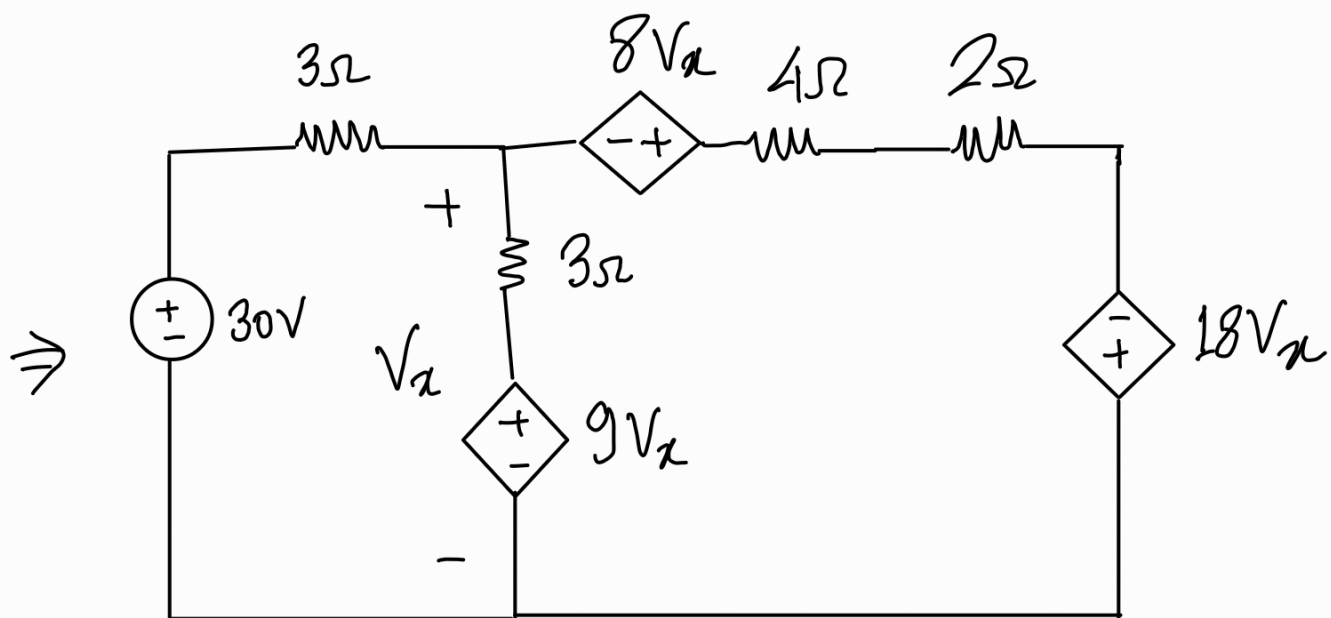
$$I_y = \frac{12 - 3I_y}{9} \Rightarrow I_y = 1A$$

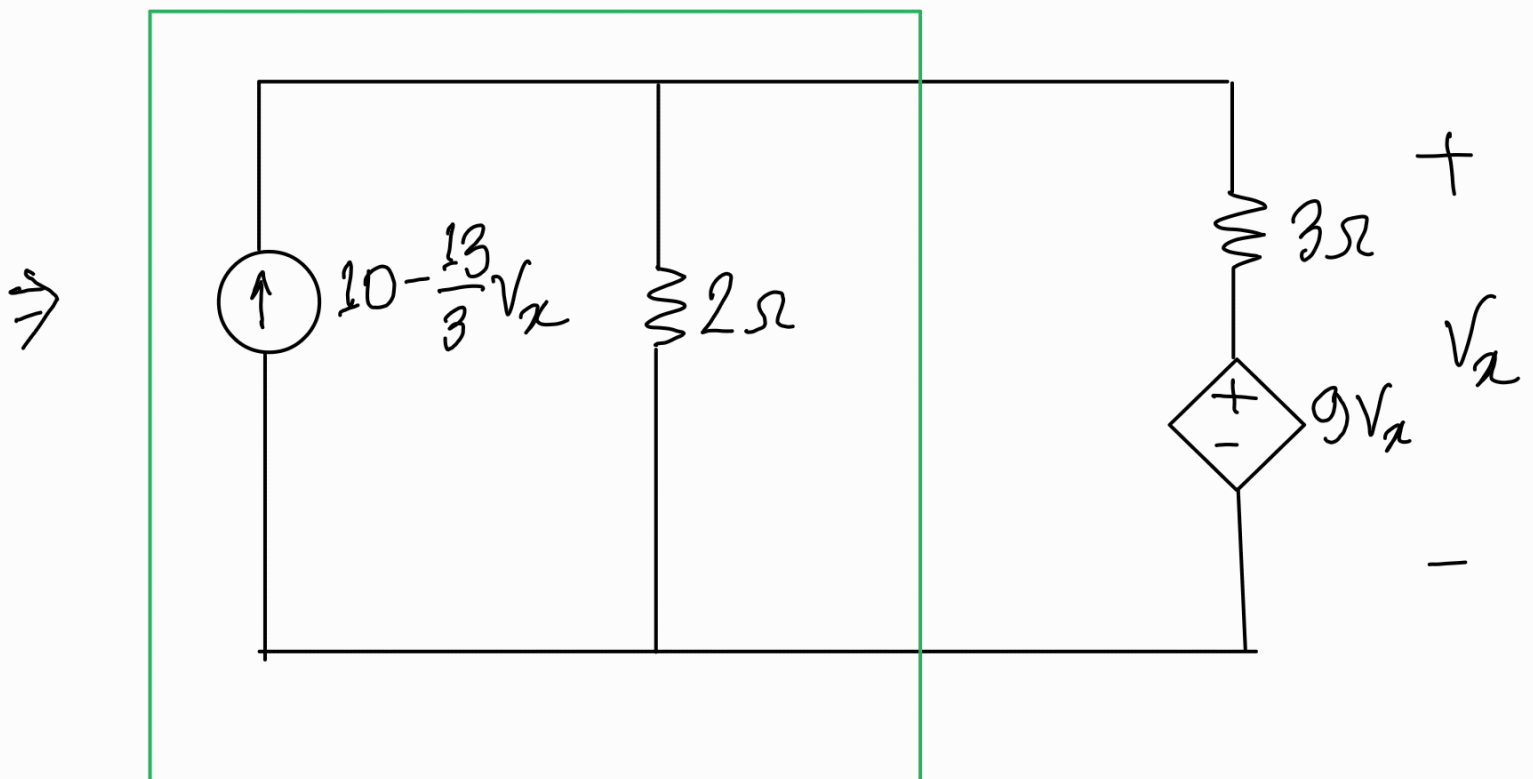
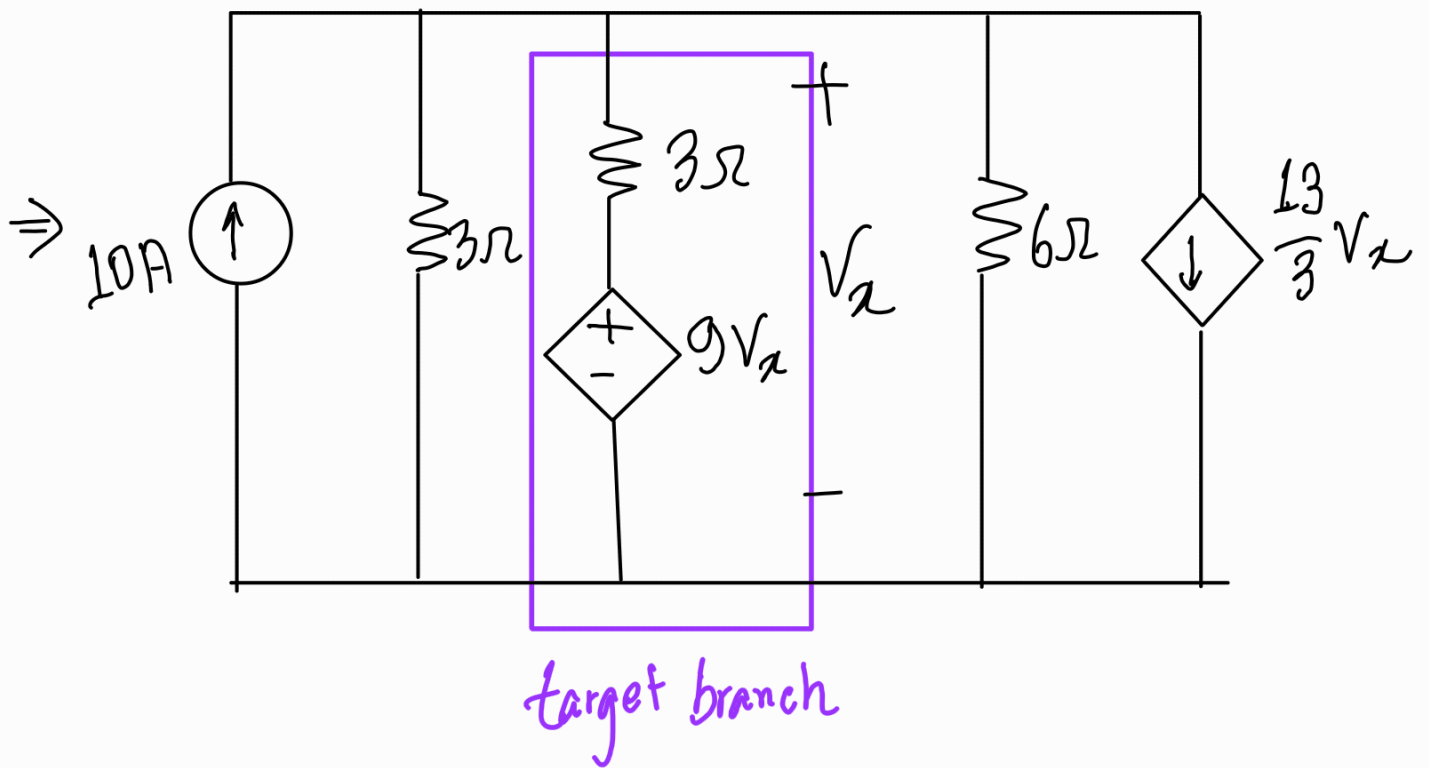
Problem 9

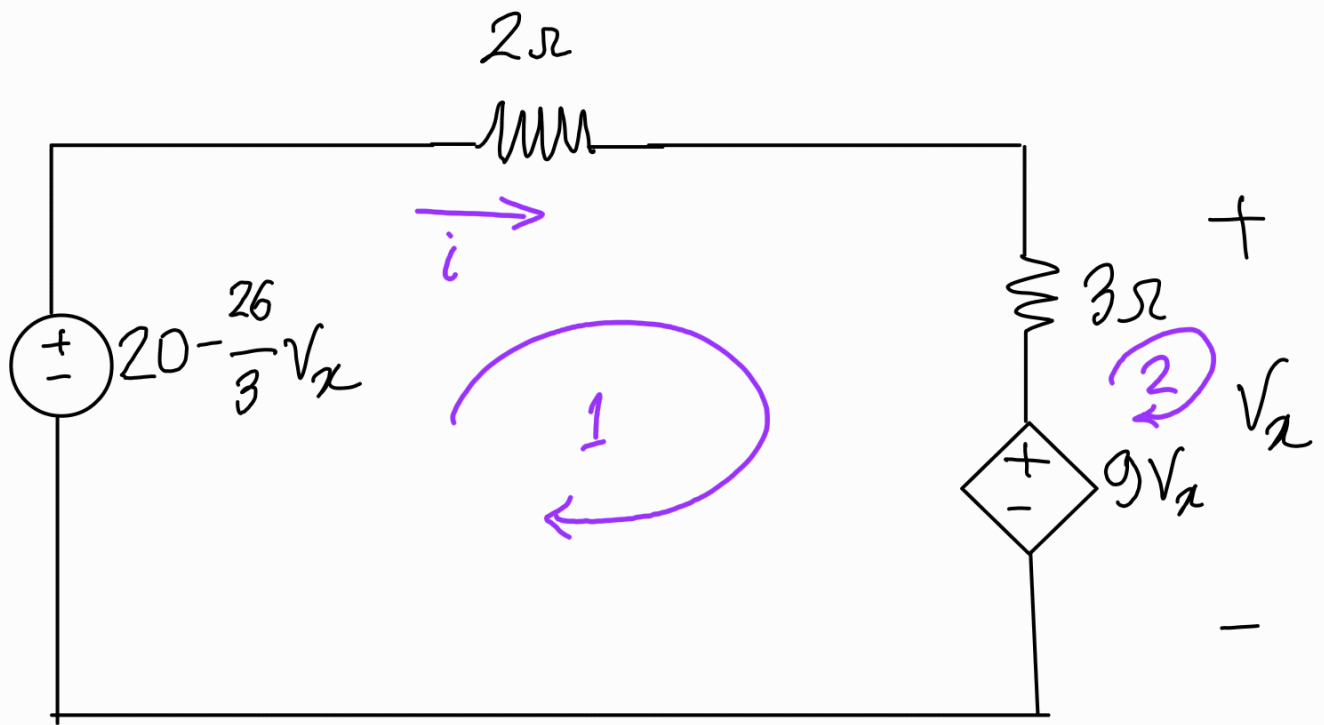


Reduce to a single loop and find V_x .









KVL @ 1 —

$$-20 + \frac{26}{3}V_x + 2i + V_x = 0$$

$$\Rightarrow 26V_x + 3V_x + 6i = 60$$

$$\Rightarrow 29V_x + 6i = 60 \text{ — (i)}$$

KVL @ 2 —

$$V_x - 9V_x - 3i = 0 \Rightarrow 8V_x + 3i = 0 \text{ — (ii)}$$

Solving (i) & (ii) —

$$V_x = 4.615 \text{ V}$$

answer

$$i = -12.31 \text{ A}$$